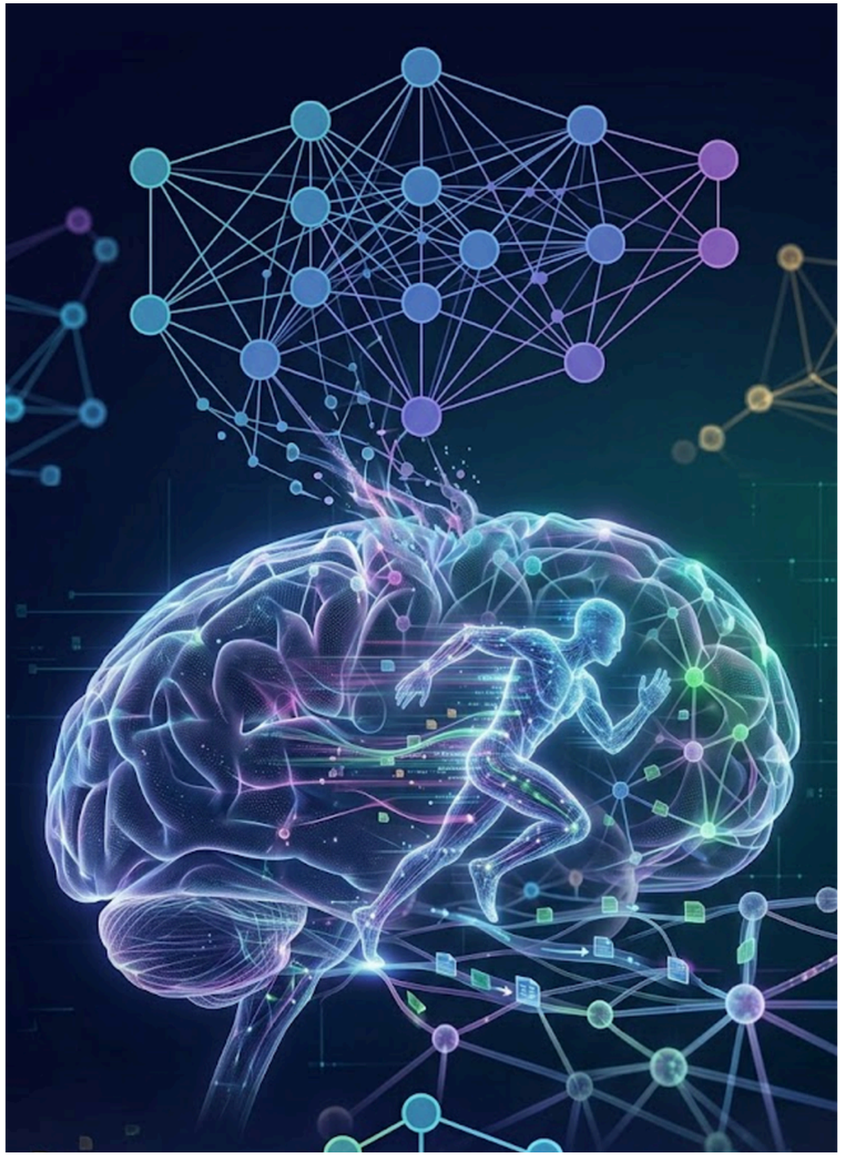
BRAC University

Leveraging Unpaired Data to Mitigate Paired Data Scarcity and Foster Generalizable Multimdal Models



Thesis Defense

Presented by : T2520826

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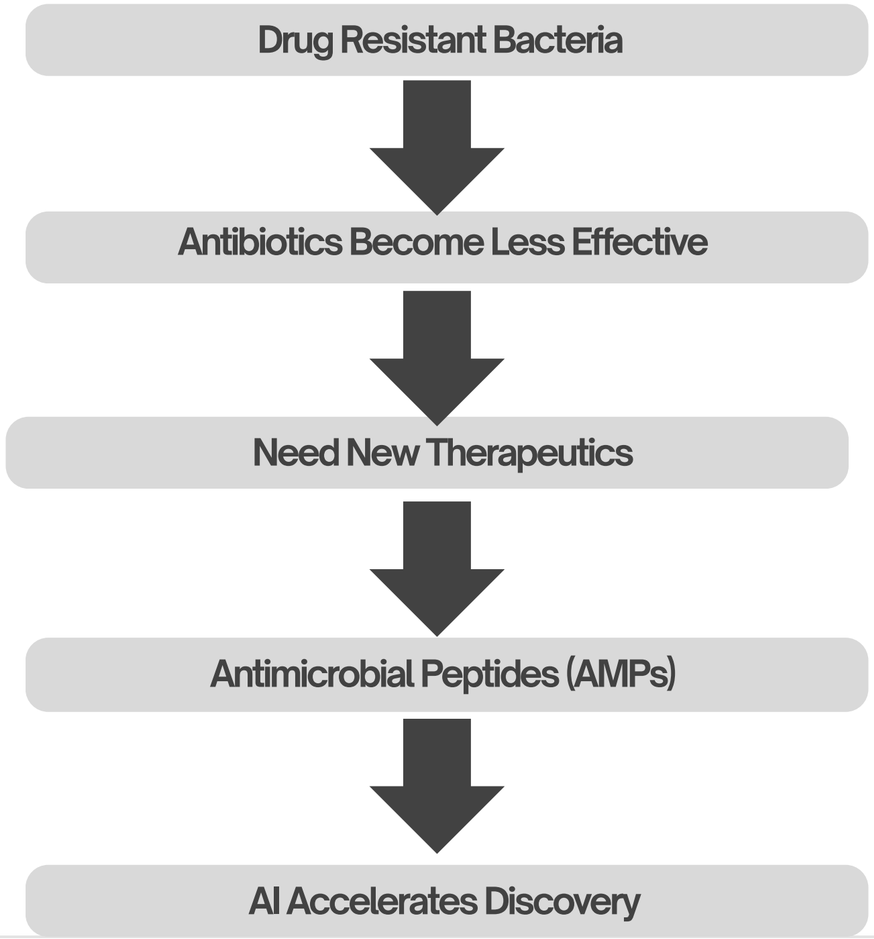
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Background and Motivation



Antimicrobial Resistance (AMR)

- One of the world's fastest-growing public health threats.
- Millions of infections are becoming difficult to treat.
- Antibiotic resistance continues to increase worldwide.

Why Are Antibiotics Failing?

- Overuse and misuse accelerate resistance.
- Development of new antibiotics is slow and costly.
- Drug-resistant bacteria continue to emerge.

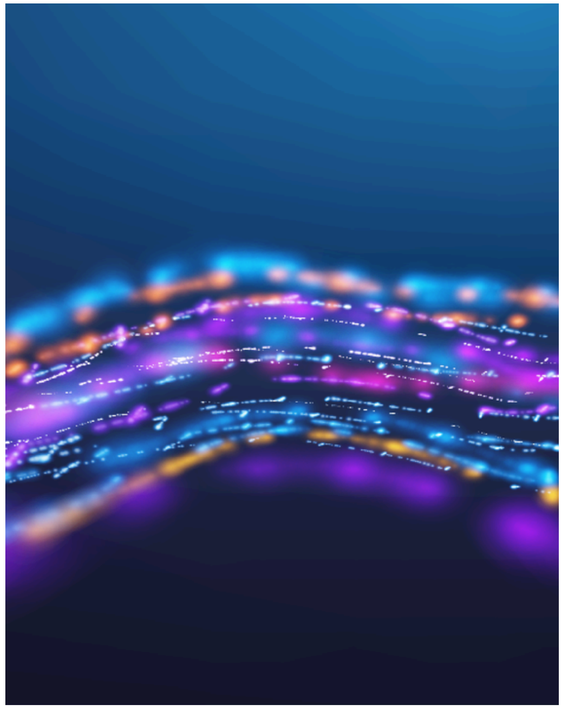
Why Antimicrobial Peptides?

- Broad-spectrum antibacterial activity.
- Multiple mechanisms reduce resistance development.
- Promising candidates for next-generation therapeutics.

Why Artificial Intelligence?

- Rapidly analyzes large biological datasets.
- Learns complex sequence–activity relationships.
- Accelerates antimicrobial peptide discovery.

Problem Statement



Existing Challenges:

Limited Species-Specific Prediction

- Most existing models predict general antimicrobial activity rather than the MIC against a specific bacterial species.

Insufficient Biological Representation

- Sequence-based models often overlook physicochemical characteristics that strongly influence peptide activity.

Biologically Inconsistent Training Data

- Many publicly available datasets combine antibacterial, antifungal, antiviral, and antiparasitic peptides, introducing biological heterogeneity and reducing prediction reliability.

Limited Prediction Reliability

- Most existing approaches provide only a single prediction without estimating prediction confidence or uncertainty.

Research Need:

Therefore, there is a need for a robust species-conditioned antibacterial MIC prediction framework that integrates rich sequence representations, biophysical knowledge, and reliable prediction strategies.

Research Objectives

Objective 1

Construct a high-quality bacteria-only dataset for species-conditioned MIC prediction.

Objective 2

Generate context-aware peptide representations using the ESM2 protein language model.

Objective 3

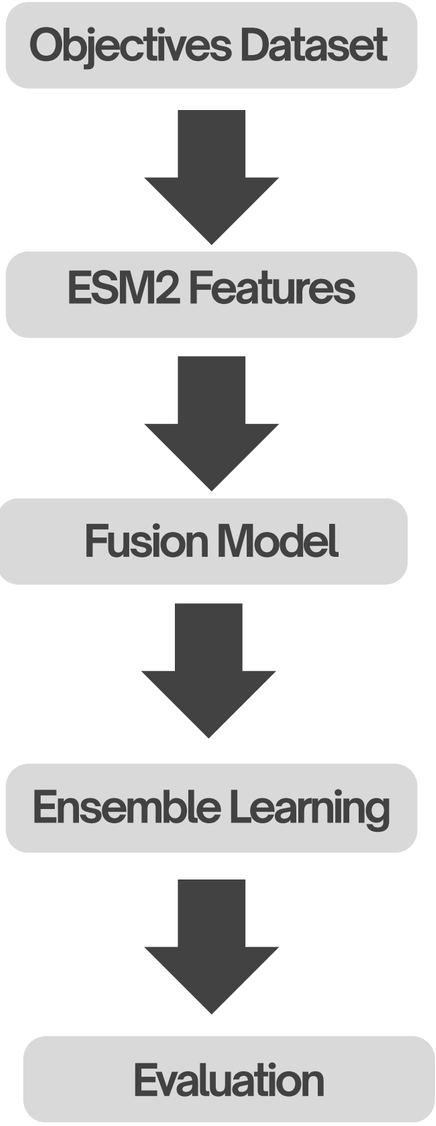
Integrate biophysical descriptors with ESM embeddings for enhanced feature representation.

Objective 4

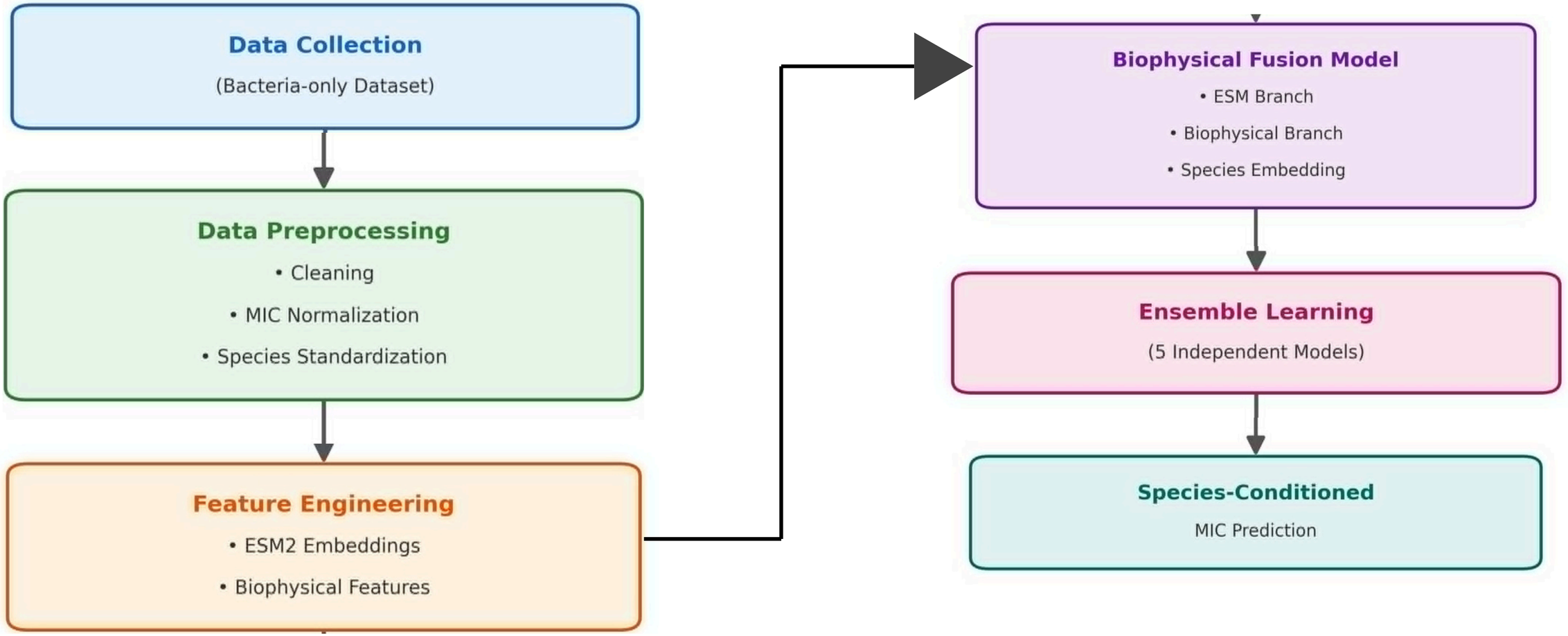
Develop a Biophysical Fusion Ensemble model for accurate antibacterial MIC prediction.

Objective 5

Evaluate the proposed framework using internal testing and external DRAMP benchmarking.



FRAMEWORK OVERVIEW



The proposed framework integrates ESM2 embeddings, biophysical descriptors, species conditioning, and ensemble learning to predict antibacterial MIC values

Dataset and Preprocessing

Data preprocessing pipeline

Raw antimicrobial activity data
36,245 records · 394 species

↓

Filter antibacterial records
keep bacterial targets only

↓

Remove duplicates & missing entries
exact + peptide–species duplicates

↓

Sequence cleaning
uppercase · 20 standard AAs · length filter

↓

MIC standardization
numeric MIC → $\log_{10}$ scale

↓

Species normalization
harmonize target-species names

↓

Train / validation / test split
70 / 15 / 15, no leakage

↓

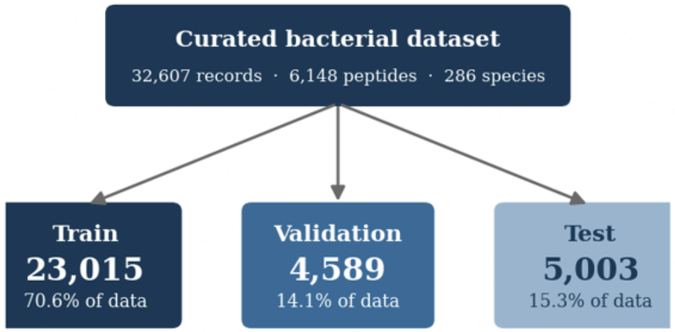
Final bacteria-only dataset

32,607	6,148	286
records	peptides	species

Final Dataset Statistics

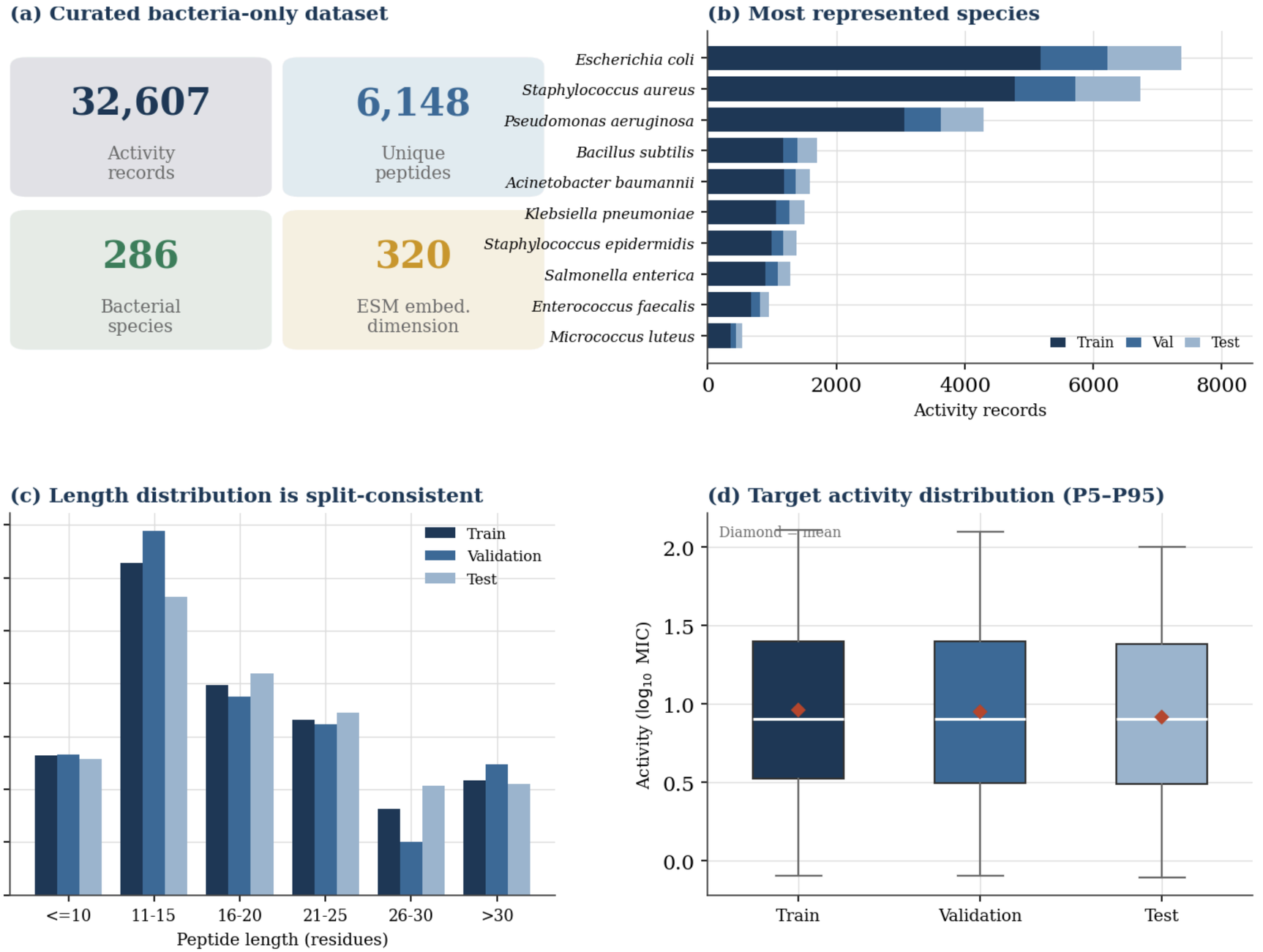
Metric	Value
Activity Records	32,607
Unique Peptides	6,148
Target Species	286
ESM Embeddings	6,148
Embedding Dimension	320

(a) Train / validation / test split

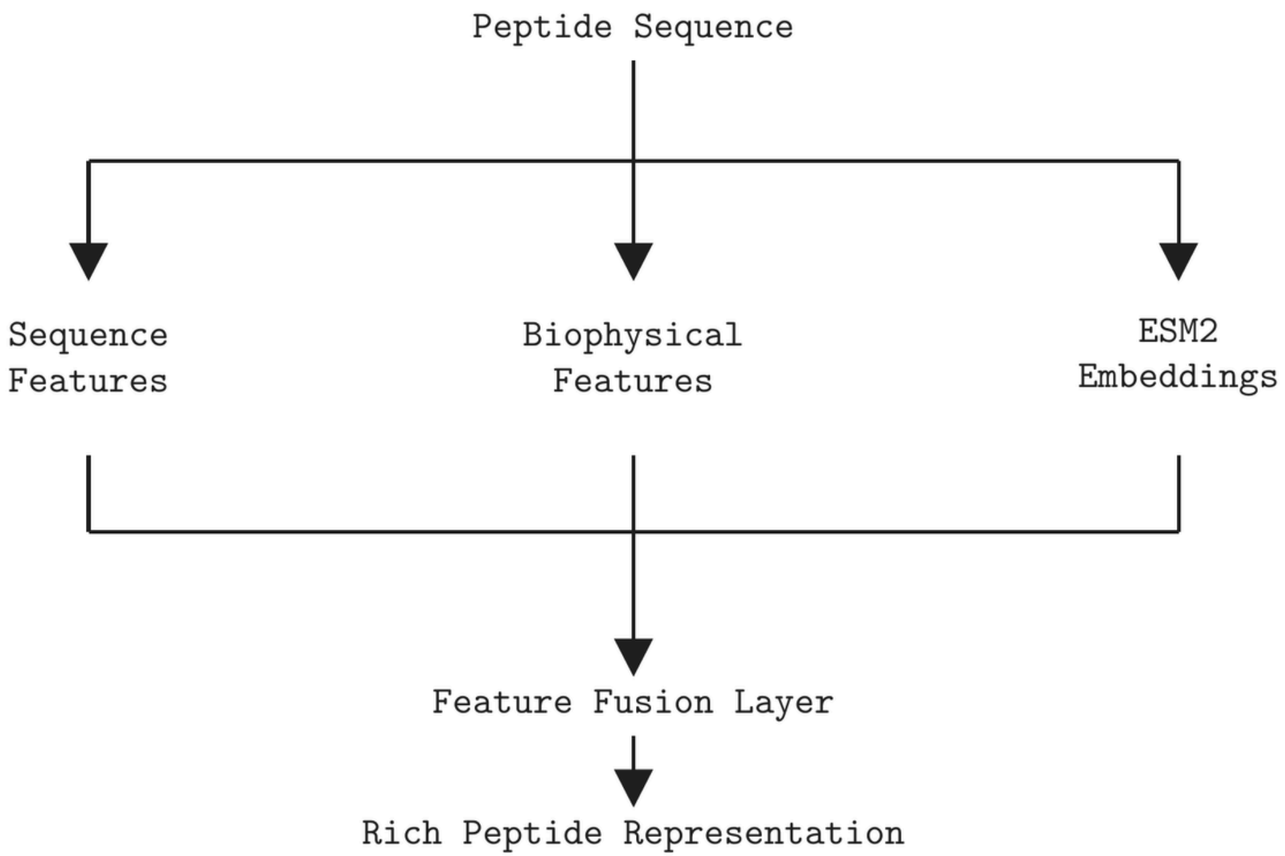


Stratified 70 / 15 / 15 split · no sequence leakage across splits

Dataset Composition



Feature Engineering

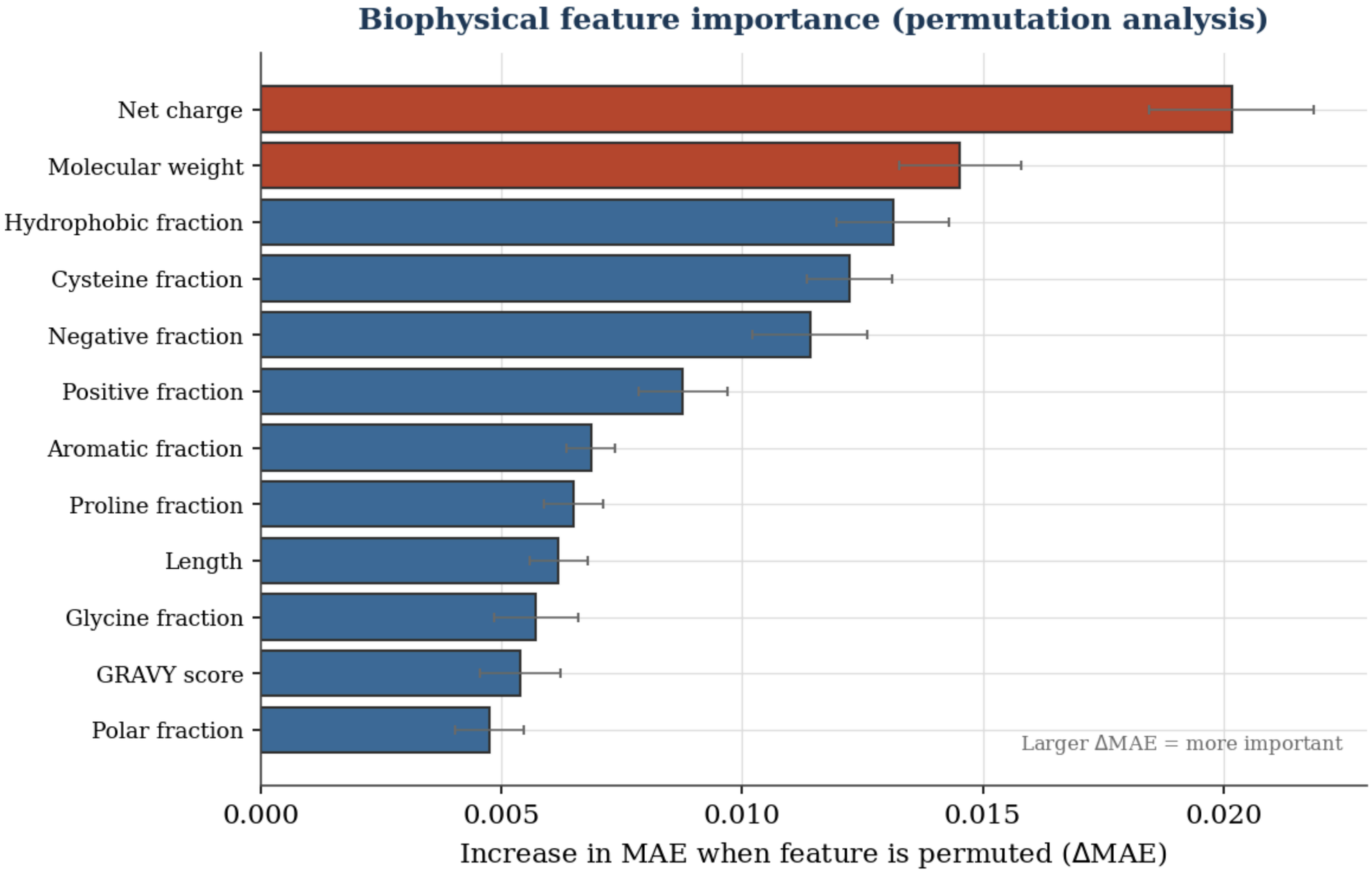


Combines sequence information, biophysical descriptors, and contextual protein embeddings to generate a comprehensive representation of antimicrobial peptides.

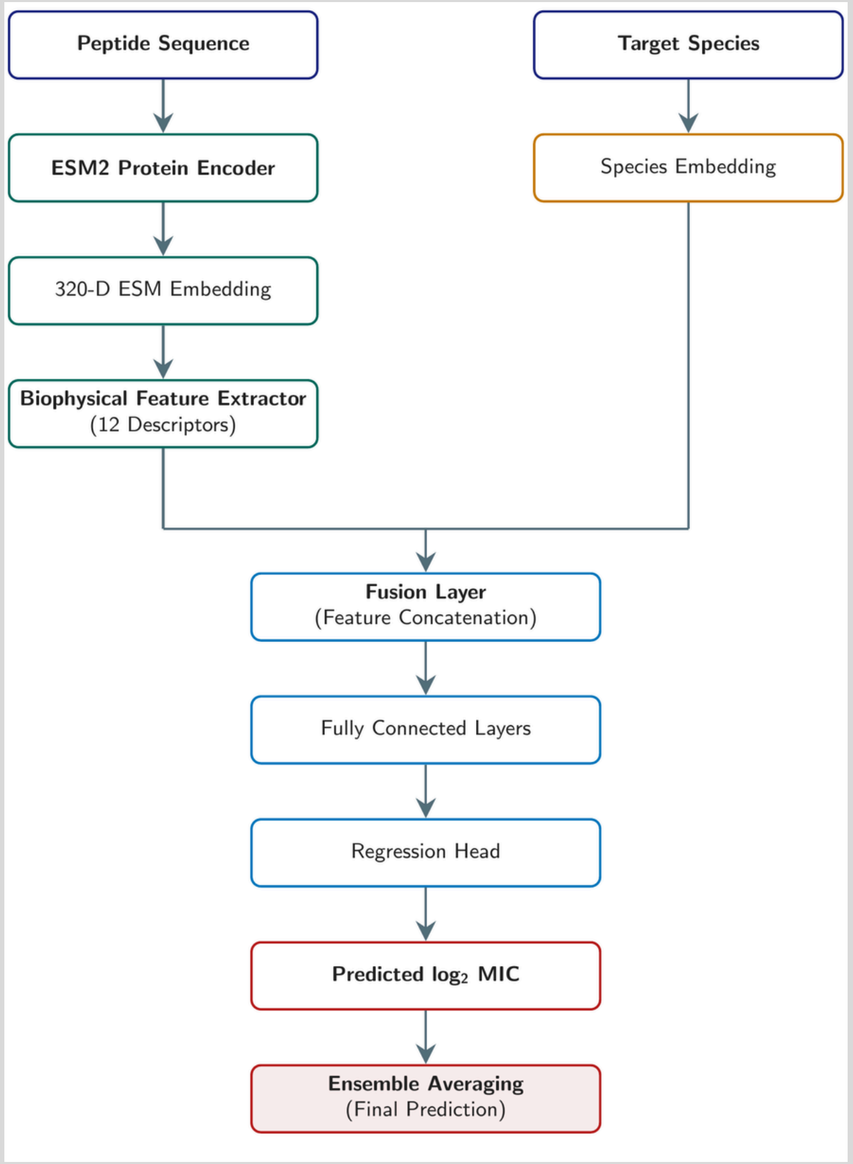
Feature Summary

Feature Type	Purpose
Sequence Features	Capture peptide amino acid composition
Biophysical Features	Describe physicochemical properties
ESM2 Embeddings	Learn contextual biological representations

Feature Importance



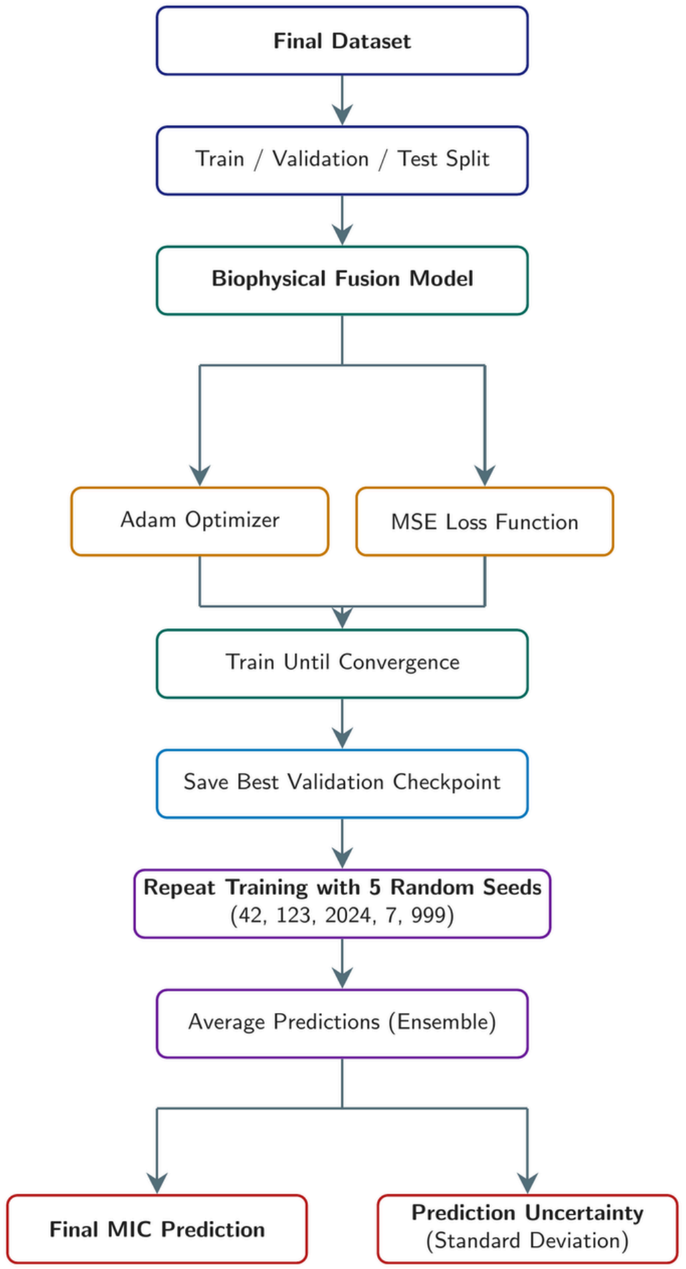
Biophysical Fusion Architecture



Component	Function
ESM Branch	Learns contextual sequence representations
Biophysical Branch	Captures physicochemical properties
Species Embedding	Encodes bacterial target information
Fusion Layer	Integrates all feature sources
Regression Head	Predicts species-conditioned MIC value

proposed architecture jointly integrates contextual sequence embeddings, biophysical descriptors, and bacterial species information into a unified regression framework for species-conditioned antibacterial MIC prediction.

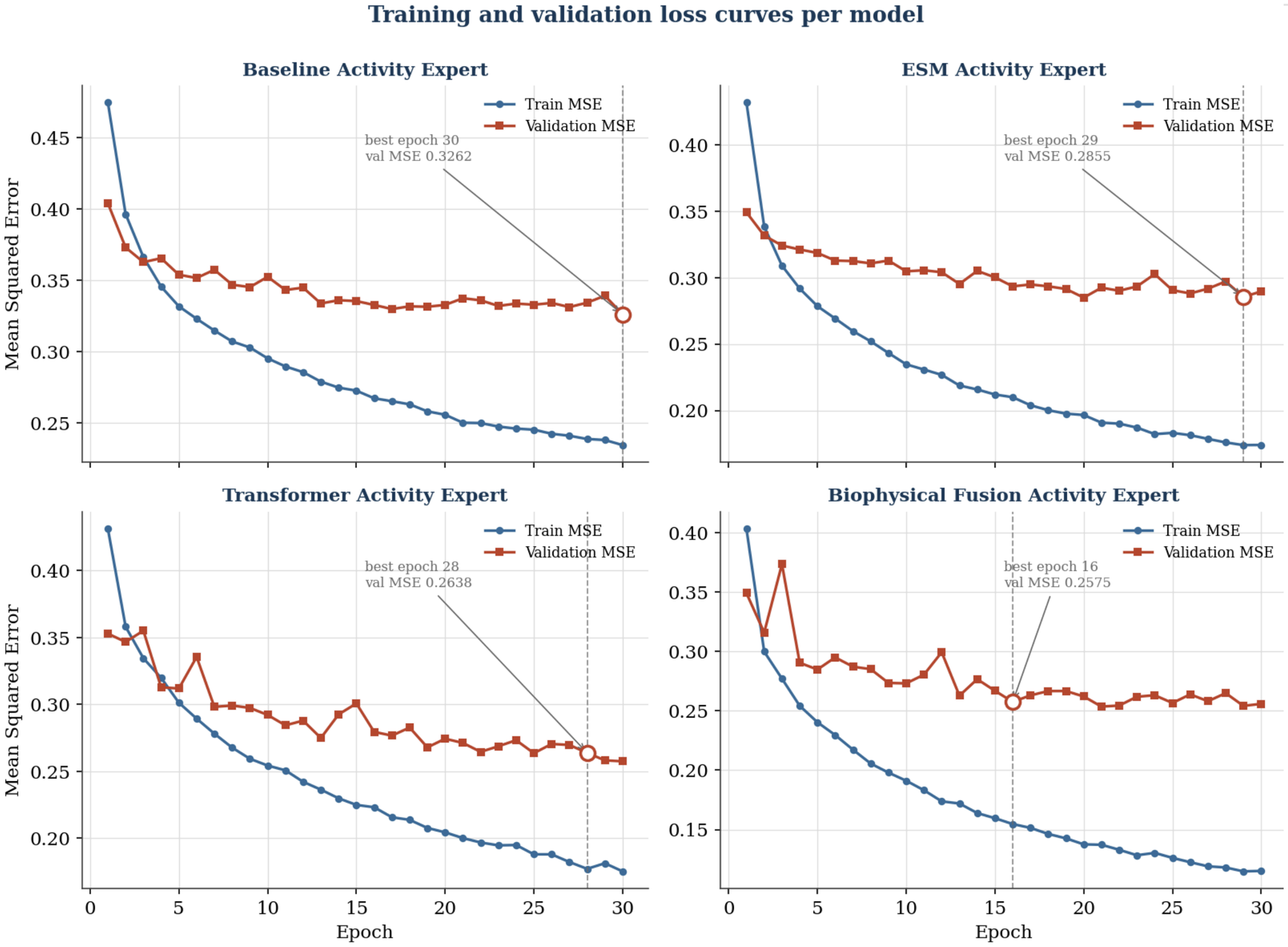
Training Procedure & Ensemble Learning

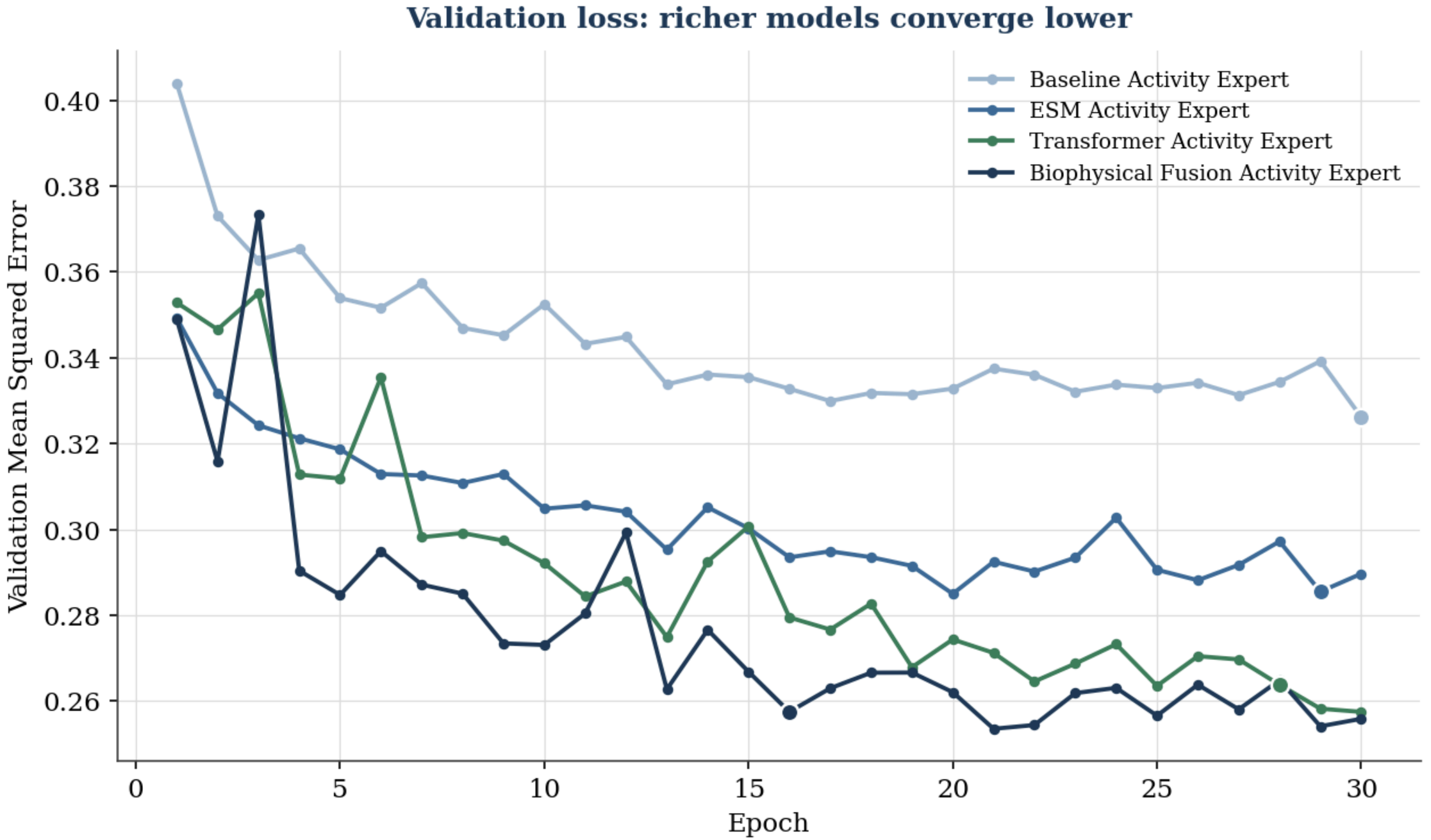


Parameter	Value
Optimizer	Adam
Loss Function	Mean Squared Error (MSE)
Model Selection	Best Validation Checkpoint
Ensemble Size	5 Models
Random Seeds	42, 123, 2024, 7, 999

Five independently trained Biophysical Fusion models were combined through ensemble averaging to improve prediction accuracy, robustness, and estimate prediction uncertainty.

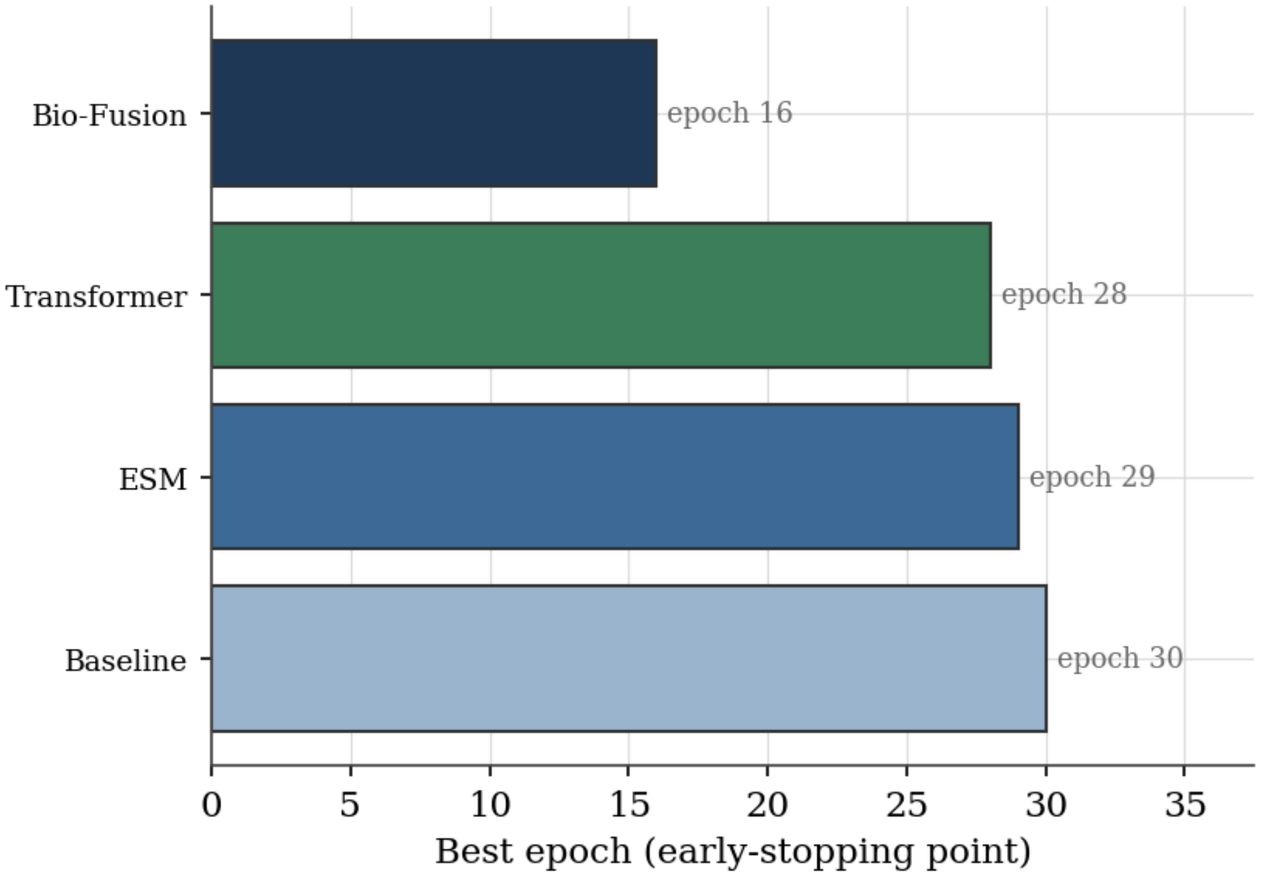
Training Curves



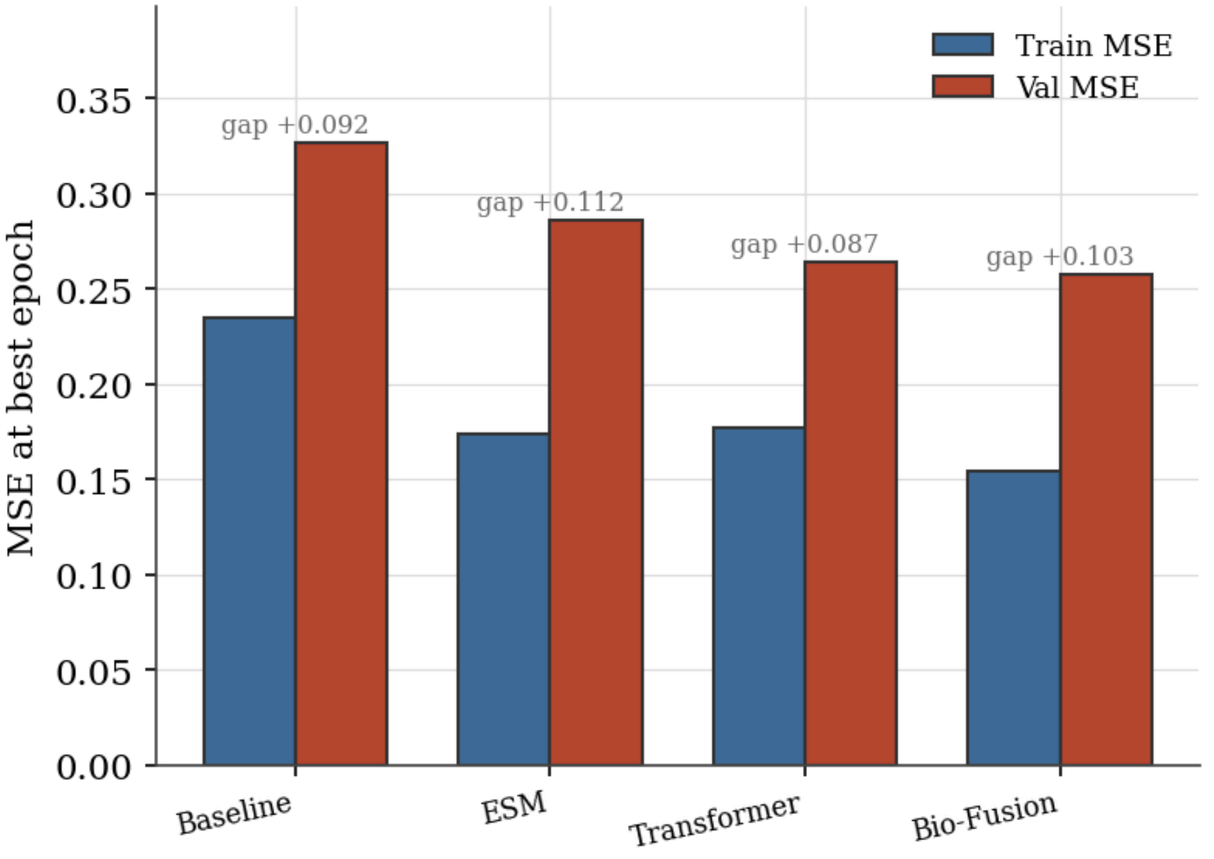

Validation loss: richer models converge lower
Baseline Activity Expert
ESM Activity Expert
Transformer Activity Expert
Biophysical Fusion Activity Expert
Validation Mean Squared Error
0.40
0.38
0.36
0.34
0.32
0.30
0.28
0.26
0
5
10
15
20
25
30
Epoch

Model training convergence and generalization

Convergence speed



Generalization gap (train vs val)



Evaluation Metrics

MAE

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Measures average prediction deviation

RMSE

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Penalizes larger prediction errors

Fold Error

$$\text{Fold Error} = 2^{\mathrm{MAE}}$$

Measures deviation in fold-change

Pearson (r)

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}$$

Measures linear correlation

Spearman (ρ)

Rank Correlation

Measures monotonic relationship between predicted and actual values

Lower MAE, RMSE, and Fold Error indicate better prediction accuracy, whereas higher Pearson and Spearman coefficients indicate stronger agreement between predicted and experimental MIC values.

Internal Performance Comparison

Model	Architecture	MAE ↓	RMSE ↓	Fold Error ↓
Baseline Activity Expert	Basic Neural Network	0.4250	0.5484	2.6605×
ESM Activity Expert	ESM2 Embeddings	0.3968	0.5234	2.4932×
Transformer Activity Expert	Transformer + ESM2	0.3910	0.5119	2.4606×
Biophysical Fusion Expert	ESM2 + Biophysical Features	0.3824	0.5030	2.4121×
Biophysical Fusion Ensemble	Fusion + Ensemble Learning	**0.3625**	**0.4774**	**2.3039**×

Legend: ↓ Lower values indicate better performance.

Key Findings

- ✓ Progressive improvement from Baseline → Ensemble
- ✓ ESM embeddings enhanced sequence representation
- ✓ Biophysical descriptors further improved prediction accuracy
- ✓ Ensemble learning achieved the lowest prediction error

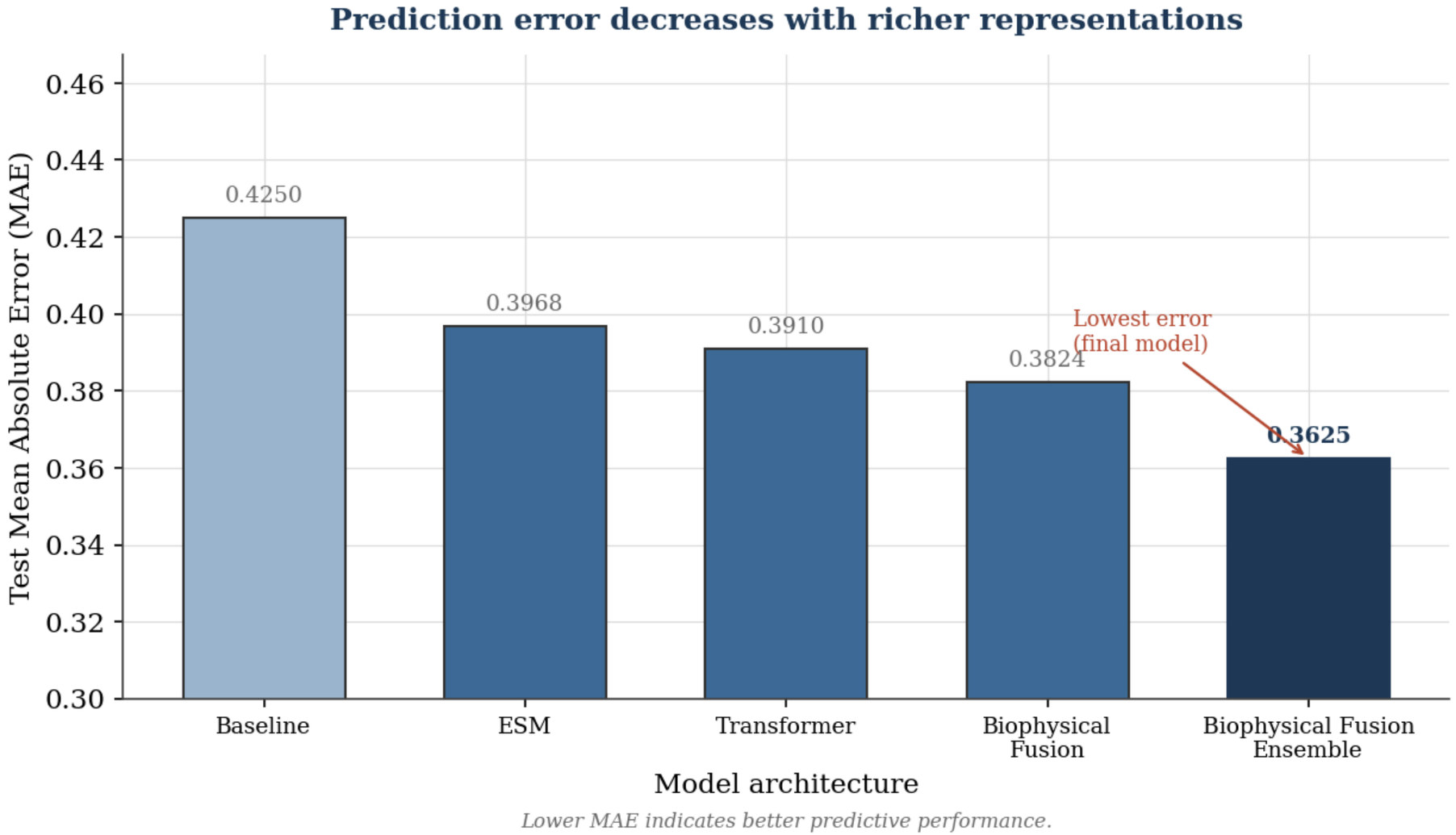
Best Model

★ **Biophysical Fusion Ensemble**

MAE = **0.3625** | RMSE = **0.4774**

Fold Error = **2.3039**×

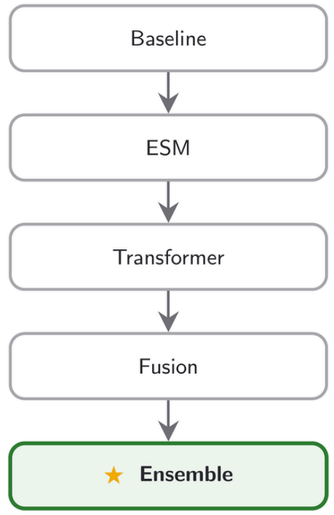
Performance Improvement



Baseline MAE : 0.4250
Ensemble MAE : 0.3625 ↓

14.7% Reduction in MAE Prediction Error

Performance Progression



Best Result

★ MAE = 0.3625

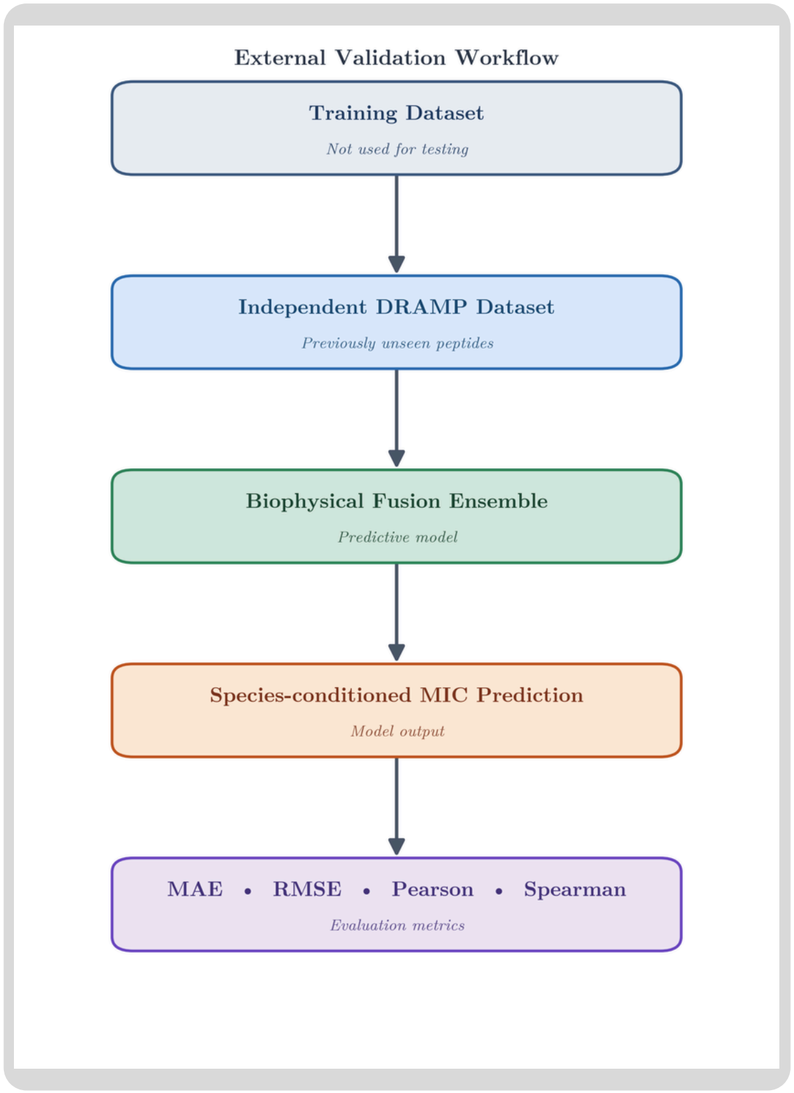
External Benchmark Evaluation

External Validation on the Independent DRAMP Benchmark Dataset

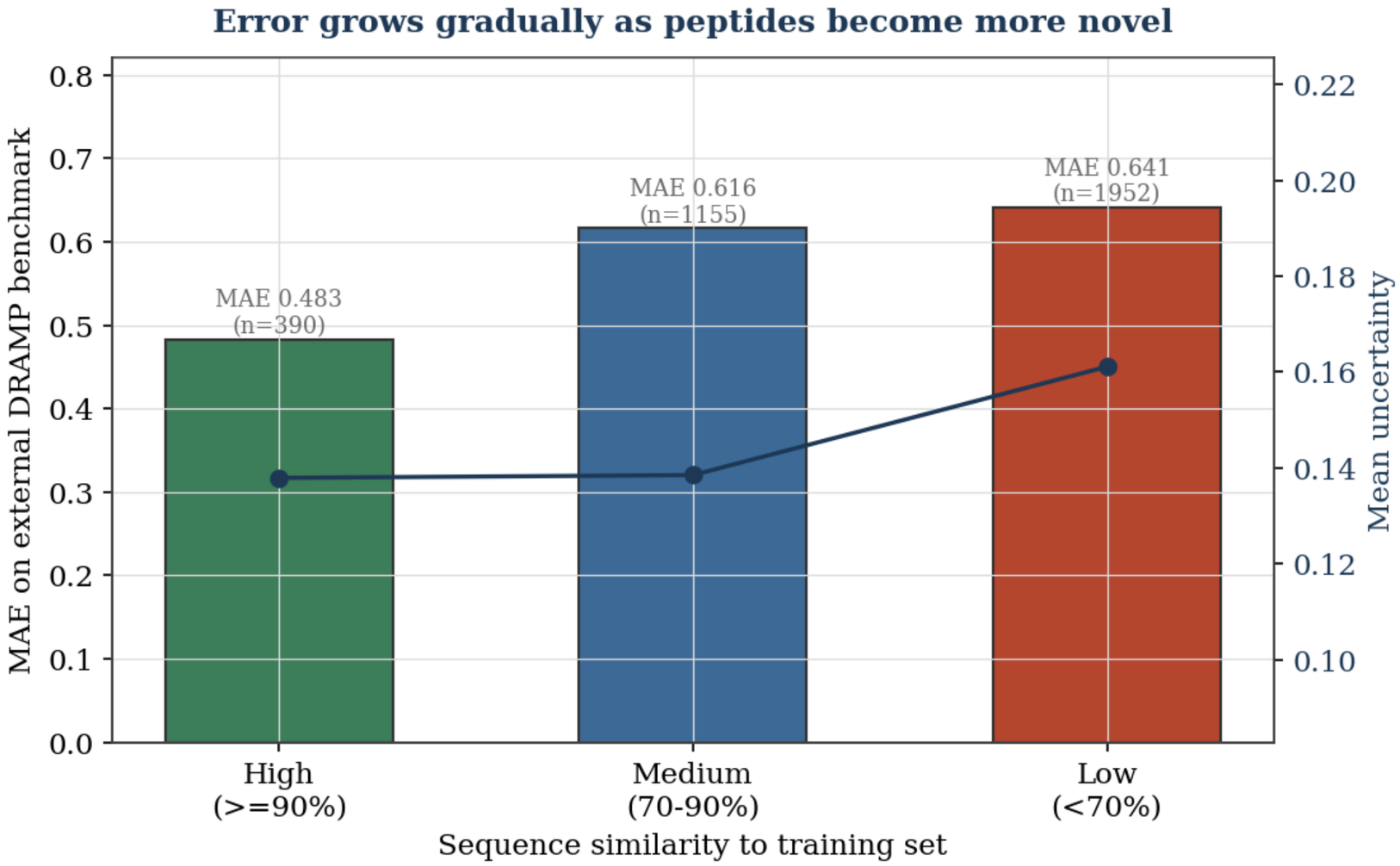
Metric	Result
External Dataset	DRAMP
Evaluation Type	Independent External Benchmark
MAE ↓	**0.6152**
RMSE ↓	**0.8514**
Pearson (r) ↑	**0.4064**
Spearman (ρ) ↑	**0.3988**

Key Findings

- Evaluated on an independent DRAMP benchmark containing previously unseen peptides.
- Achieved MAE = 0.6152 and RMSE = 0.8514.
- Moderate Pearson (0.4064) and Spearman (0.3988) correlations indicate meaningful agreement with experimental MIC values.
- Results demonstrate successful generalization beyond the training dataset.



Novelty Analysis



The proposed Biophysical Fusion Ensemble maintained stable predictive performance across increasing peptide novelty, indicating that it learned biologically meaningful sequence–activity relationships rather than memorizing the training data.

Candidate generation and filtering funnel

Generated candidates
1,000 peptides

↓ 10% retained

Activity top-k
100 peptides

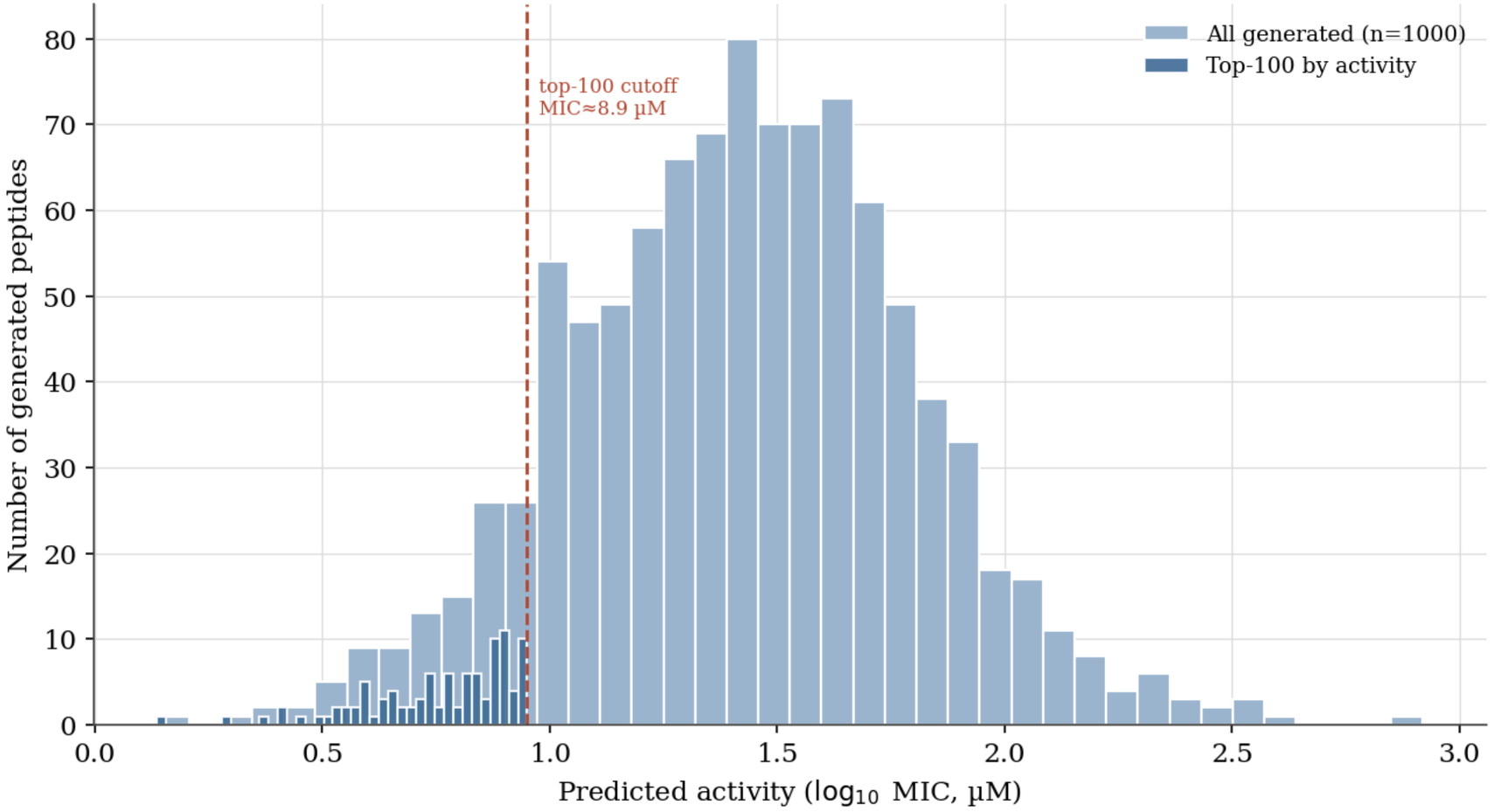
↓ 25% retained

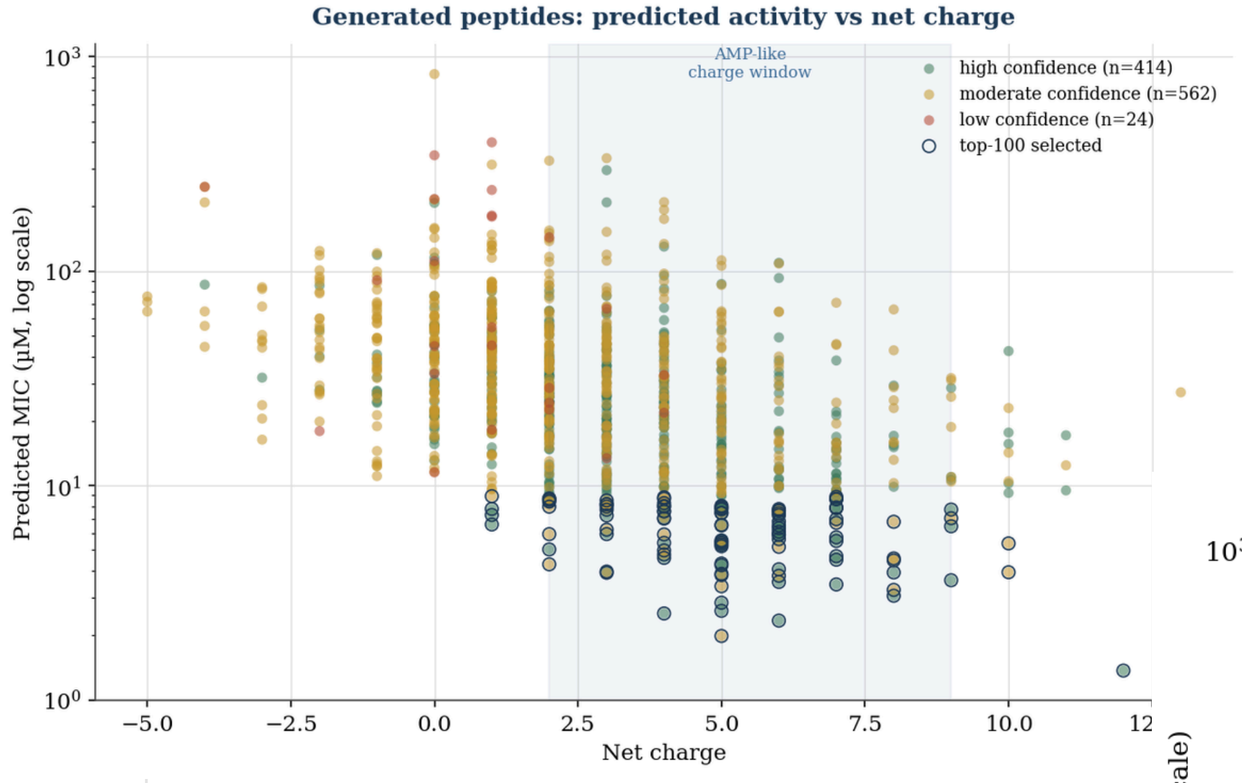
Diversity-filtered
25 peptides

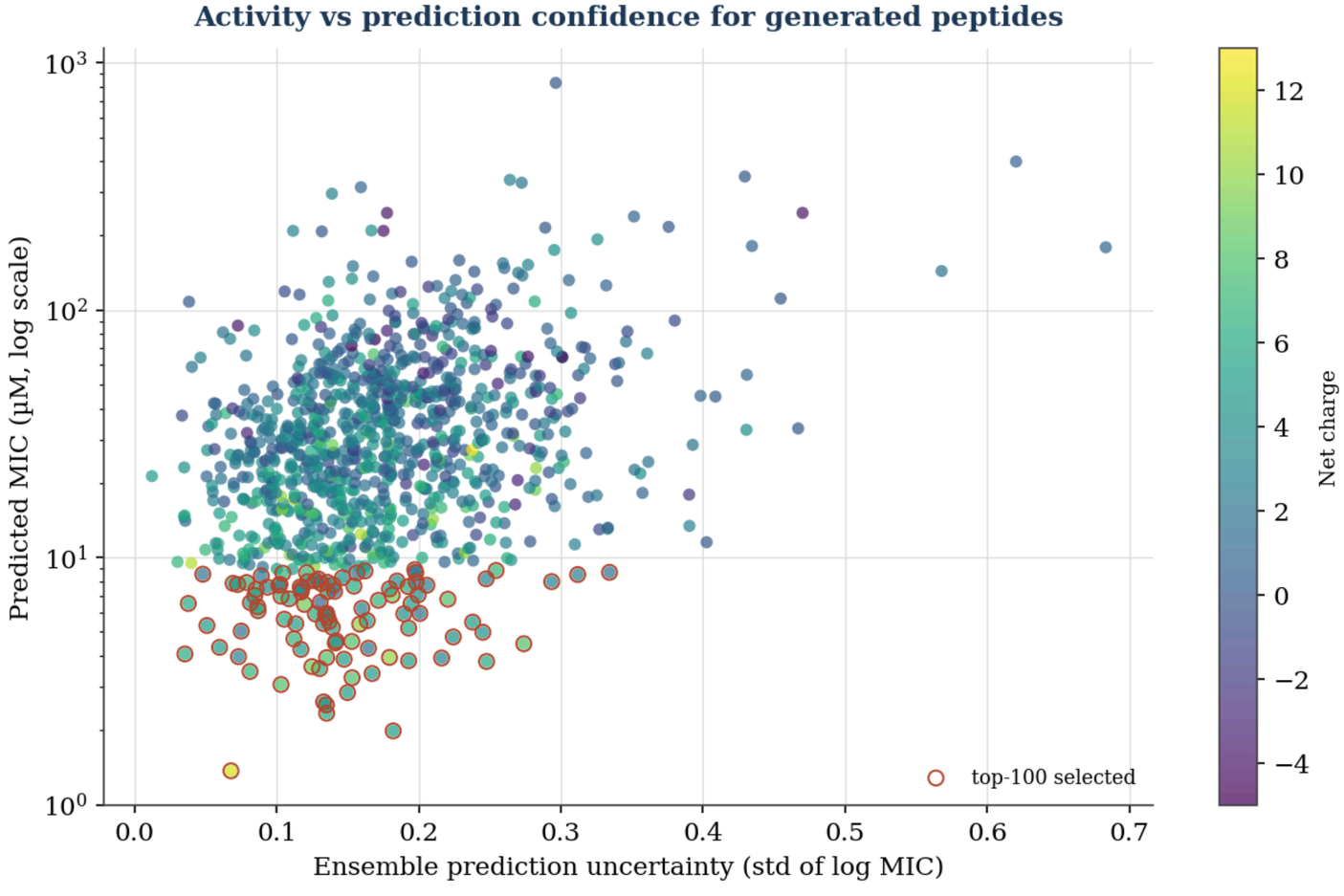
↓ 56% retained

Final candidates
14 peptides

Predicted activity distribution of generated peptides




Generated peptides: predicted activity vs net charge
AMP-like
charge window
high confidence (n=414)
moderate confidence (n=562)
low confidence (n=24)
top-100 selected
Predicted MIC (µM, log scale)
Net charge

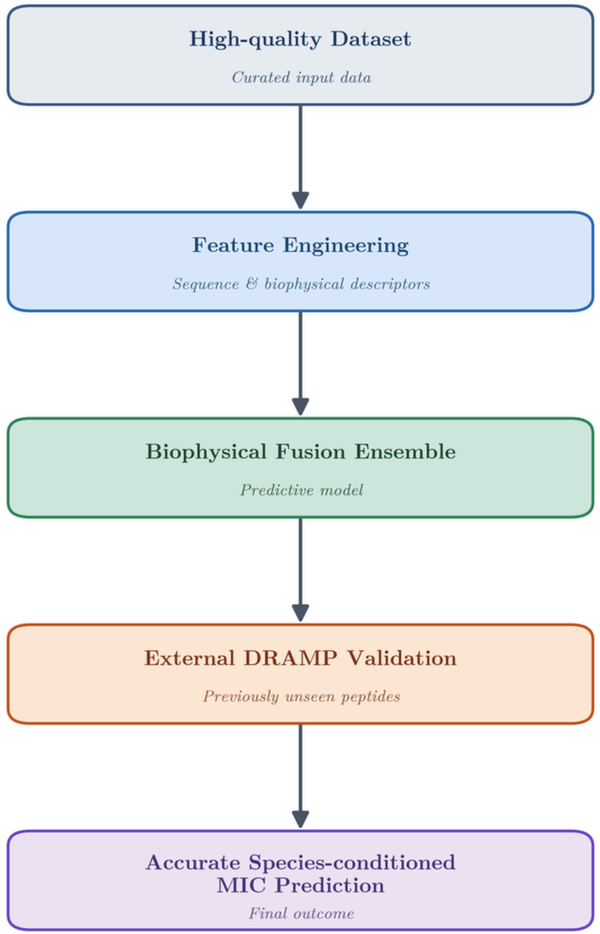

Activity vs prediction confidence for generated peptides
top-100 selected
Predicted MIC (µM, log scale)
Ensemble prediction uncertainty (std of log MIC)
Net charge

Final generated antimicrobial peptide candidates (E. coli target)

Tier	Sequence	Pred. MIC (μM)	Uncertainty	Confidence	Max train identity	Ensemble stability
Tier 1	KKRRWFWLEARPRRWLFWTERS	2.35	±0.135	high	4.5%	highly stable
Tier 1	IWITTFLDKQLGLPTVGNWAHLKKGGLKHI	2.60	±0.133	high	6.7%	stable
Tier 1	KGKLWKNIWKLAKMFNRWMAEFQS	2.84	±0.150	high	8.3%	stable
Tier 1	KKKLRKRIRRWRSEFVDIAVVI	3.46	±0.081	high	8.0%	highly stable
Tier 1	WAKAVLIKAAALKRGLRDRKRAKK	3.62	±0.125	high	20.8%	stable
Tier 1	KTQRRNWIKQWIVWFRRDFRI	4.07	±0.035	high	13.6%	stable
Tier 2	LLRLRLGWKLVKFFCNKRGWY	3.55	±0.130	high	19.0%	unstable
Tier 2	GLLKMLKKLDWVLIKWSLEKILRFKMQG	3.88	±0.147	high	14.3%	unstable
Tier 2	RTKDFTPQRYIKAILKKYKVDMNWI	4.25	±0.117	high	16.0%	unstable
Tier 2	IRSLKLRLRILGRWAKFQLKA	4.51	±0.141	high	4.8%	unstable
Tier 2	WKWRYTWSLIFAGQHHHQIVSHKLN	4.68	±0.112	high	0.0%	unstable
Tier 3	LLWRLIKLVRKVLPKAHVEAKEKGKARAP	3.26	±0.153	moderate	13.8%	unstable
Tier 3	KTKWGRWWAVRCGKTYGKA	3.80	±0.248	moderate	10.5%	unstable
Tier 3	GKWGSLLKKLLSRLAGSAAKRELIRSMLLGPD	3.82	±0.193	moderate	18.8%	unstable

Conclusion

- Species-conditioned MIC prediction framework
- Biophysical Fusion Ensemble achieved the best internal performance
- Dataset refinement improved accuracy
- External DRAMP benchmark confirmed model generalization
- Supports AI-assisted antimicrobial peptide discovery



The proposed framework provides accurate, robust, and species-conditioned antibacterial MIC prediction.

Existing Limitations And Future Research Scopes

Limitations

- Limited Dataset Size and Generalizability
- Computational Requirements
- No Experimental Validations
- Limited Scope of Biological Property Prediction

Future Research Scopes

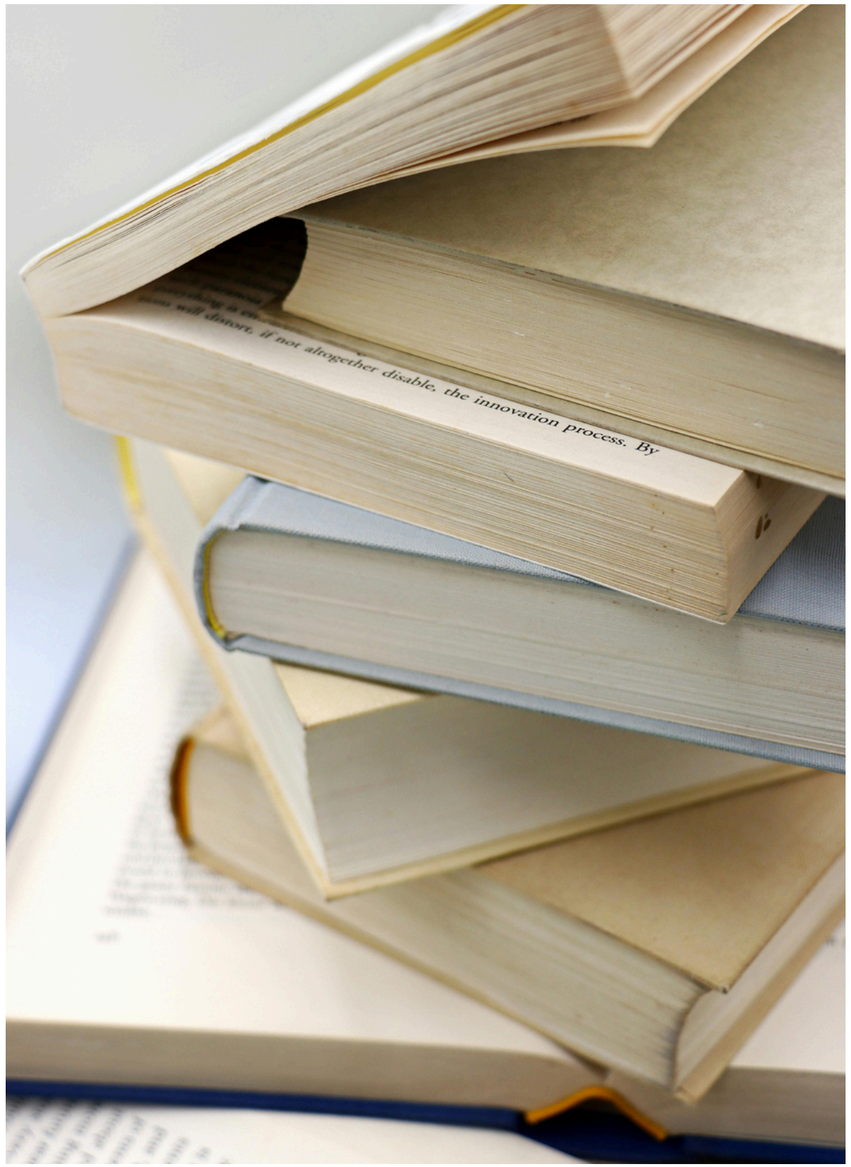
- Larger, Higher Quality, Taxonomically Diverse Datasets
- Future models could use Structural Information
- Advancing Ensemble and Multi-Task Learning Strategies
- Experimental validation and Generative Peptide Design
- Clinical Translation and Continuous Learning

The proposed framework can be extended into a complete AI-driven antimicrobial peptide discovery platform by integrating generative design, multi-property prediction, and experimental validation, thereby accelerating the development of next-generation antimicrobial therapeutics.

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Thank You

Please ask questions if you have any!



Presented by :
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